Integrated Fog Visibility Estimation and Evolution Forecasting

A unified framework combining atmospheric optics, CNN calibration, interpretable visual features, TOPSIS, ARIMA, Random Forest, XGBoost, and meteorological validation

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Core empirical results

Synthetic CNN test accuracy	92.46%
Macro F1	0.9248
Best proxy forecasting model	ARIMA(0,1,1)
Primary limitation	No synchronized real-road visibility labels for the 100-image proxy set

Abstract

How much fog is visible in a road image, and can that information help predict what happens next? We examine these questions with a CNN, five visual indicators, and three forecasting models. The available data do not all have the same labels, so we keep four experiments separate. Synthetic transmittance images have known extinction coefficients and support supervised CNN testing. A set of 100 real fog images supports feature extraction and relative severity ranking, but has no synchronized visibility measurements. A separate 100-frame highway sequence allows us to examine changes in contrast. AMOS observations provide measured meteorological optical range (MOR) for testing meteorological estimates and short-term forecasts across two events. On 4,575 balanced synthetic test images, the CNN reaches 92.46% accuracy and Macro F1 = 0.9248. For the real-image experiment, Brightness, Contrast, Saturation, Dark Channel, and Edge Density are combined using entropy weights, a provisional AHP prior, and TOPSIS. The rankings remain stable under the tested weight changes. On the 100-image proxy sequence, ARIMA(0,1,1) has lower hold-out MAE (16.3759 km^{-1}) and RMSE (19.5360 km^{-1}) than Random Forest and XGBoost under the same 80/20 split. However, all three have negative R^2 and about 70% directional accuracy. The AMOS tests also expose weaknesses: a very close training fit can fail on another fog event, and persistence outperforms the tested temporal Ridge model. These results support the individual modeling steps more strongly than an end-to-end road forecast. Connecting them to physical visibility will require consecutive road images, reliable timestamps, camera information, and independent visibility measurements.

Keywords: fog visibility; CNN; extinction coefficient; visual features; entropy weighting; AHP; TOPSIS; ARIMA; Random Forest; XGBoost; AMOS; time-series forecasting

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1. Introduction and Literature Review

Fog makes road markings and distant objects harder to distinguish because light is scattered and attenuated before reaching the camera. For a driver, the current condition matters, but so does whether it is getting worse. We therefore ask both how to estimate the present fog state and how to predict its change. A classifier alone answers only part of that problem.

1.1 Problem and Research Objective

Our study links four tasks: relate meteorological observations to visibility, estimate fog from images, assemble an image-based attenuation sequence, and predict whether conditions intensify or dissipate. Extinction coefficient β gives these tasks a shared physical quantity when calibration is available. For images without field labels, we instead use an attenuation proxy. Keeping that distinction visible is necessary before discussing safety thresholds.

A difficulty runs through the project: the easiest data to label are not the data needed to test road performance. The synthetic images have exact β labels, while the real fog images do not have simultaneous meter readings. Their contrast and color changes can still be measured, but a numerical score does not become physical visibility just because it is expressed in meters. Similarly, the order of image files does not establish the elapsed time between observations. These gaps determine how far we can interpret each experiment.

1.2 Literature Review

The optical starting point comes from work on imaging in bad weather. Nayar and Narasimhan [1] describe the contributions of direct transmission and airlight; Tan [3] shows how a single degraded image can carry useful information about fog. This helps explain why brightness, contrast, and scene depth matter in our image branch. It does not tell us how quickly a fog event will change, which requires observations over time.

Road-camera studies bring the image model closer to the quantity of interest. Hautière et al. [2] estimate visibility using road structure and Koschmieder-type contrast attenuation. Xiao et al. [7] combine structural, spectral, and depth-related information, while Yang et al. [8] use RGB, dark-channel, and edge information in a coarse-to-fine network. These studies motivate our use of learned image features alongside simple indicators whose contributions can be inspected. The remaining question is how to calibrate such outputs on our own data.

Forecasting adds another requirement: the observations must describe a sequence through time. Miao et al. [5] examine sequence models for short-term fog prediction. Penov and Guerova [9] use meteorological variables for visibility estimation, and Cornejo-Bueno et al. [6] study low-visibility motorway events. The nowcasting work in [11] also motivates comparison with persistence or climatological baselines. We take this as a practical testing rule: estimating the current state and predicting the next state need separate evaluation, even when both are used in the same system.

Synthetic fog offers controlled labels when real measurements are scarce. Sakaridis, Dai, and Van Gool [4] demonstrate its value for learning, but the gap between generated and real fog remains relevant here. Our synthetic transmittance experiment tests whether the CNN can distinguish its labeled inputs. It does not establish that the same accuracy holds for highway MOR measurements.

1.3 Research Gap and Contributions

The link between an image estimate and a forecast is the part we focus on. A camera score must be clear enough to interpret and consistent enough to use as a time-series input. We use the CNN to learn density classes, TOPSIS to summarize five visible image characteristics, and ARIMA, Random Forest, and XGBoost to compare forecasting approaches. At present, these are connected modeling components; calibration on common road measurements is still needed to test the complete chain.

- A CNN test with known synthetic β labels and a confusion matrix that shows which density classes are confused.
- A five-indicator severity ranking, with entropy and AHP weights, TOPSIS, and checks of sensitivity to weight choices.
- A fusion equation that converts CNN β from m^{-1} to km^{-1} before combining outputs, while keeping calibration unresolved.
- A comparison of ARIMA, Random Forest, and XGBoost on the same ordered training and hold-out observations.
- Separate AMOS and highway tests that show where the image and forecasting results do, and do not, transfer.

1.4 Data Sources and Evidence Boundaries

Dataset / stream	Scale	Supervision	Valid use
Synthetic transmittance test set	4,575 test images; 1,525 per β class	Exact β class from filename / synthetic generation	CNN classification performance in synthetic domain
Real Fog image set	100 images (page01_001.png ... page10_100.png)	No synchronized field visibility label	Five features, entropy/AHP/TOPSIS, proxy attenuation sequence
Highway screenshot sequence	100 time-ordered frames	No meter label; timestamps partially recoverable	Relative contrast trend only; no absolute MOR claim
AMOS meteorological observations	2,878 valid minute records across two event periods	Measured MOR plus T/RH/wind/pressure	Cross-event meteorological and temporal validation

Table 1. Data streams are deliberately kept separate because their labels and validation strength differ.

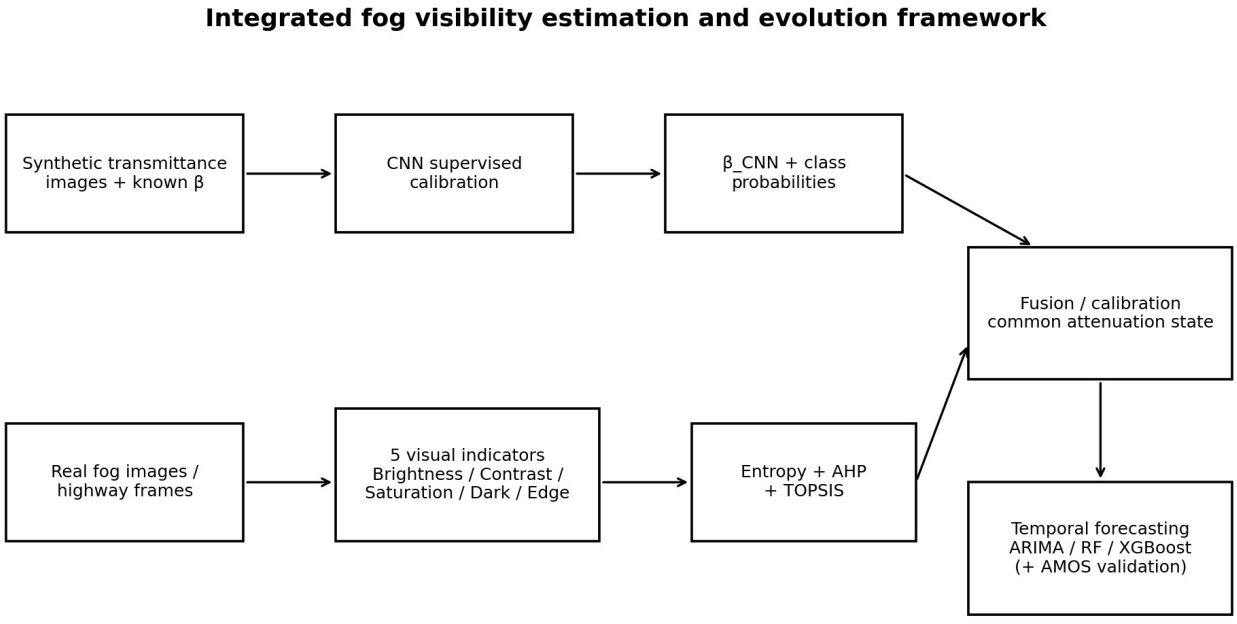


Figure 1. Integrated framework. Perception, interpretable severity, calibration, and temporal forecasting are linked without treating all datasets as interchangeable training samples.

2. Methodology

2.1 Atmospheric Image Formation and Visibility Conventions

Assuming spatially homogeneous attenuation, the recorded image contains two contributions: scene light that passes through the fog and airlight scattered toward the camera:

$$I_t(x) = J(x)\,\tau_t(x) + A_t\,[1 - \tau_t(x)]$$

$$\tau_t(x) = \exp[-\beta_t\, d(x)]$$

The extinction coefficient β controls attenuation with distance. Setting the threshold transmittance to 5% gives the following visibility conversion:

$$V_{5\%} = -\ln(0.05) / \beta = 2.995732 / \beta$$

With the 5% convention, β = 0.005, 0.010, and 0.020 m^{-1} correspond to about 599.15, 299.57, and 149.79 m. The earlier real-image proxy uses a different conversion, based on the 2% Koschmieder constant 3.912. We retain that scale when reporting the existing forecasting results so the numbers remain reproducible. The two conversions cannot be treated as interchangeable; a calibrated combined model would need one agreed field definition.

2.2 Synthetic Fog-Density CNN

The input is a grayscale transmittance PNG resized to 32 × 64. Three stride-2 convolutional blocks expand the channels from 1 to 8, 16, and 24. Adaptive average pooling, dropout 0.1, and a 24→3 linear head complete the classifier, which has 4,803 trainable parameters. The recorded training run uses cross-entropy loss and AdamW for five epochs. Its best validation accuracy occurs at epoch 5.

Stage	Output	Operation
Input	1 × 32 × 64	Normalized transmittance image
Conv 1	8 × 16 × 32	3×3, stride 2, padding 1, ReLU
Conv 2	16 × 8 × 16	3×3, stride 2, padding 1, ReLU
Conv 3	24 × 4 × 8	3×3, stride 2, padding 1, ReLU
Adaptive AvgPool	24 × 1 × 1	Global spatial aggregation
Classifier	3 logits	Dropout 0.1 + Linear(24,3)

Table 2. Compact CNN architecture used for the synthetic transmittance classification benchmark.

A hard label discards the probabilities of the other classes. Keeping those probabilities allows a weighted average of the three nominal β values:

$$\beta_{CNN,t} = 0.005\,P_{1,t} + 0.010\,P_{2,t} + 0.020\,P_{3,t} \quad (m^{-1})$$

$$\beta_{CNN,t}\,(km^{-1}) = 1000\,\beta_{CNN,t}$$

Multiplying by 1000 expresses CNN β in km^{-1}, the unit used by the TOPSIS proxy. Matching units is necessary for fusion, although it does not resolve the calibration gap.

2.3 CNN Evaluation

The test contains 4,575 synthetic images, with 1,525 in each class. We report accuracy, precision, recall, F1, cross-entropy, and the confusion matrix. For an additional interpretation of the class errors, predictions are mapped to nominal visibilities of 600/300/150 m and an MAE is calculated. The 30-image random check samples the same test domain; it is a quick check of predictions, not a new external test.

2.4 Five Real-Image Visual Indicators

Indicator	Operational definition	Interpretation	TOPSIS direction
Brightness	Mean image intensity	Higher value treated as more severe	Benefit
Contrast	Global structural distinguishability	Fog tends to reduce contrast	Cost
Saturation	Color preservation	Fog tends to desaturate	Cost
Dark Channel	Local minimum-channel statistic	Scattering raises the dark-channel statistic	Benefit
Edge Density	Fraction of visible edge pixels	Fog suppresses fine structure	Cost

Table 3. Five interpretable indicators extracted from the 100 real fog images.

The indicators are standardized over the 100 images. Their directions define our severity score: for example, lower contrast is treated as more severe fog. This is a modeling choice, since scene content and camera exposure can also change these features. The indicators make the score easier to inspect, but they do not supply missing visibility labels.

2.5 Entropy + AHP + TOPSIS

We compare entropy weights with a provisional AHP prior (CR = 0.026). The combined scheme gives each equal influence by setting $\alpha = 0.5$; this is a chosen balance, not a fitted parameter:

$$w_{j,comb} = \alpha\, w_{j,AHP} + (1 - \alpha)\, w_{j,Entropy}$$

Indicator	Entropy	AHP	Combined
Brightness	0.0214	0.1814	0.1014
Contrast	0.0485	0.3188	0.1836
Saturation	0.0128	0.1282	0.0705
Dark Channel	0.0217	0.2748	0.1482
Edge Density	0.8956	0.0968	0.4962

Table 4. Indicator weights. Edge Density dominates the entropy weights but is moderated by the AHP prior in the combined scheme.

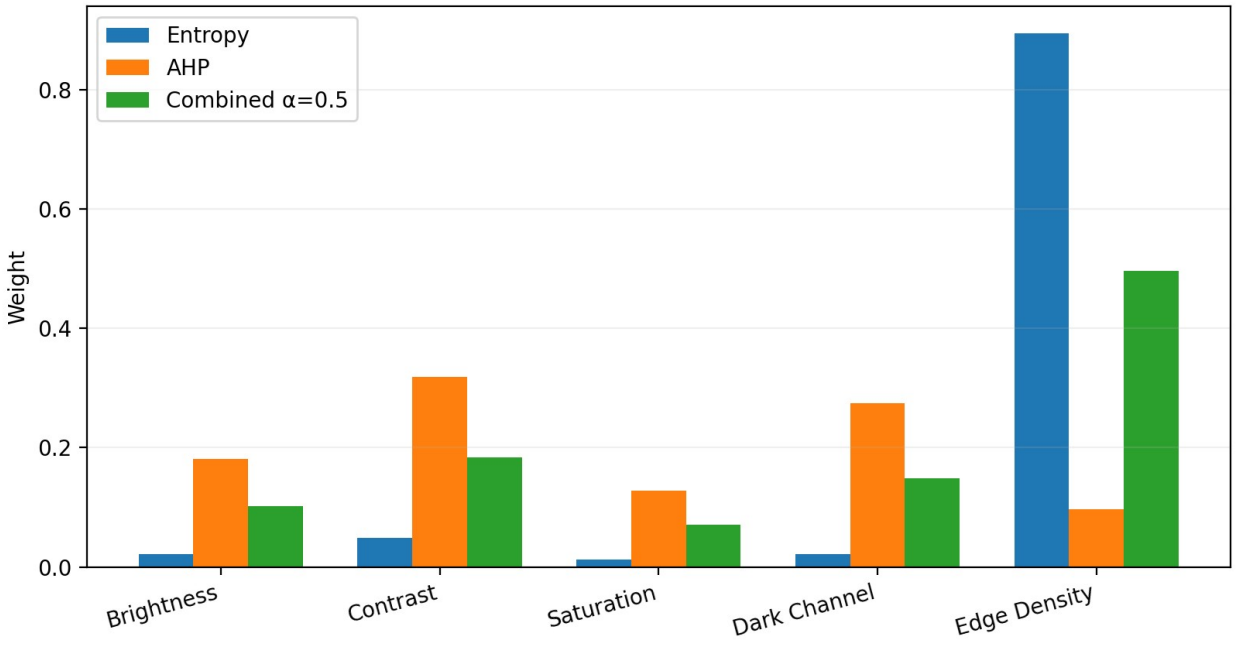


Figure 2. Entropy, provisional AHP, and combined weights.

TOPSIS compares each weighted, normalized observation with the severe-fog ideal and its opposite. Their relative distances define the score C_i:

$$C_i = D_i^- / (D_i^+ + D_i^-)$$

Higher C_i indicates closer agreement with the chosen severe-fog ideal. We perturb individual weights by ±20% and vary α from 0 to 1. If the rankings changed sharply, interpretation would depend heavily on the selected weights.

2.6 CNN/TOPSIS Fusion and Calibration

The CNN output has synthetic β supervision, whereas TOPSIS has no field visibility labels. The existing proxy mapping supplies a numerical scale for the latter:

$$\beta_{TOPSIS,t} = 3.912 + 74.328\, C_t \quad (km^{-1}, \text{legacy proxy scale})$$

After unit conversion, a provisional weighted combination can be expressed as:

$$\beta_{state,t} = \lambda\, \beta_{CNN,t}^{(km^{-1})} + (1 - \lambda)\, \beta_{TOPSIS,t}$$

This equation specifies how the outputs might be combined; it does not validate the resulting coefficient. The branches still differ in calibration convention, and the real images have no simultaneous visibility-meter labels. To estimate λ for road use, both outputs would need calibration against the same field β or MOR measurements, with separate real-road observations reserved for testing.

2.7 ARIMA Temporal Model

We fit ARIMA on observations 1–80 and reserve 81–100 for evaluation. The candidate orders cover p = 0–3, d = 0–2, and q = 0–3; AIC selects ARIMA(0,1,1). Keeping the supplied order avoids mixing later observations into training. However, we cannot independently verify that these 100 images form a continuous recording. The split therefore tests prediction along the supplied sequence, with no established physical time interval between its points.

2.8 Random Forest and XGBoost Forecasting

Random Forest and XGBoost predict the next proxy β from its previous three observations. They use the same first-80 / last-20 split as ARIMA, without random shuffling, and random seed 42. We compare MAE, RMSE, R^2, and directional accuracy. This gives the nonlinear models the same target observations and a short history from which to learn.

$$x_t = [\beta_{t-1}, \beta_{t-2}, \beta_{t-3}] \;\rightarrow\; \beta_t$$

The comparison asks whether three lagged values contain nonlinear predictive structure that the tree ensembles can use. Their greater flexibility is useful only if it improves the hold-out predictions.

2.9 AMOS Meteorological and Temporal Validation

AMOS supplies an independent test with measured MOR, temperature, relative humidity, wind speed, and pressure. Minute aggregation and quality control leave 2,878 valid observations across two event periods. We fit on the first period and test on the second. This is a harder check than holding out nearby points from the same event, because the fitted relationship must transfer to another period.

For meteorological estimation, Ridge and Random Forest predict log attenuation using standardized T, RH, wind, and pressure. Temporal Ridge uses current and lagged log visibility plus current meteorological variables to forecast 5 and 15 minutes ahead. At both horizons we also keep current visibility unchanged as the persistence forecast. It is a simple baseline, but one that a useful short-term model needs to beat.

2.10 Highway Relative-Visibility Analysis

For the separate highway recording, we measure relative change in 100 ordered frames. Each of five fixed road ROIs gives a contrast statistic based on its 90th and 10th grayscale percentiles. We take the median across ROIs and apply a causal seven-frame median smoother. Without camera geometry or known clear-scene distances, the output remains dimensionless. It can describe an image trend, but cannot supply an absolute visibility distance.

$$c_{t,j} = [Q90(I_{t,j}) - Q10(I_{t,j})] / [Q90(I_{t,j}) + Q10(I_{t,j}) + \varepsilon]$$

3. Results

3.1 CNN Results

The CNN gets 4,230 of 4,575 synthetic test images correct: 92.46% accuracy and Macro F1 = 0.9248. Test cross-entropy is 0.23637. Across the five recorded epochs, validation accuracy rises from 83.47% to 93.00%, while validation loss falls from 0.4857 to 0.2331.

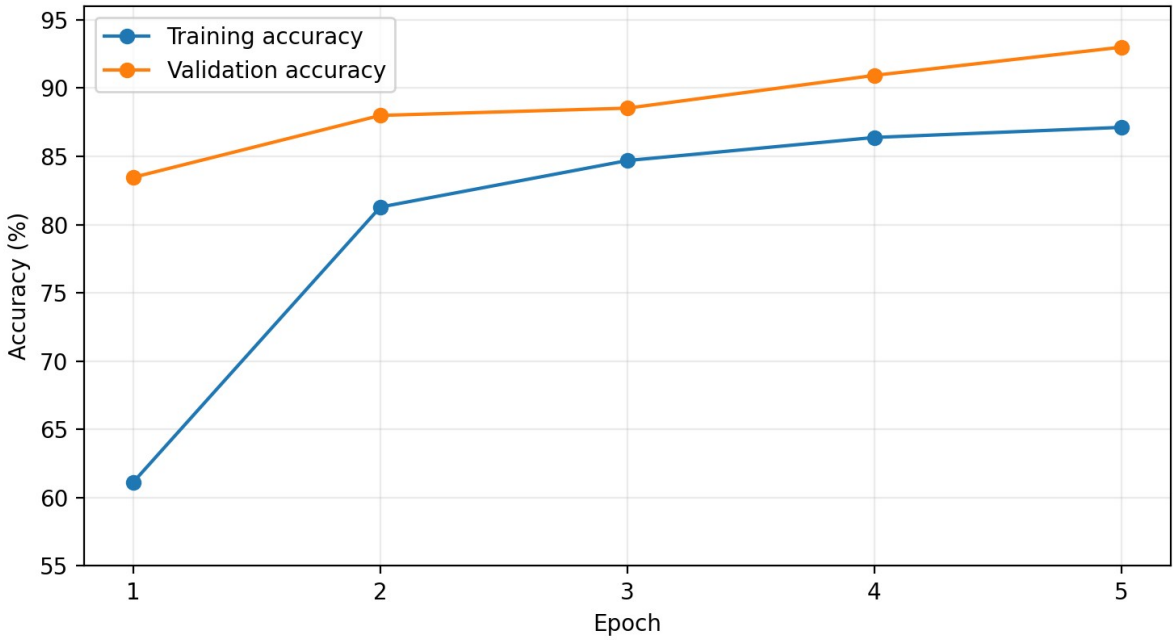


Figure 3. Training and validation accuracy over the five recorded epochs.

β class	Precision	Recall	F1
0.005	0.9358	0.8984	0.9167
0.010	0.8715	0.9075	0.8892
0.020	0.9691	0.9679	0.9685
Macro / overall	0.9255	0.9246	0.9248

Table 5. Classwise CNN performance on the balanced synthetic test set.

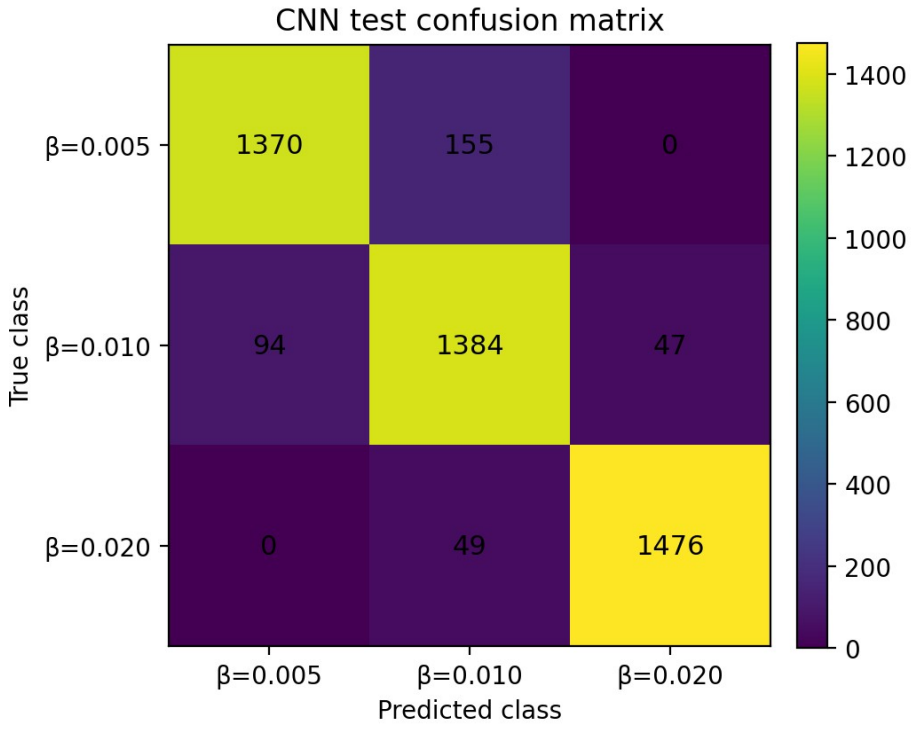


Figure 4. Confusion matrix. All observed errors are between adjacent β classes; no 0.005 ↔ 0.020 direct confusion occurs in this test set.

Mapping classes to 600, 300, and 150 m gives a discrete visibility MAE of 19.48 m. The errors also have a direction that matters: 49 of 1,525 strong-fog images (3.21%) are assigned β = 0.010 instead of 0.020, shifting nominal visibility from about 150 m to 300 m. The random check gets 28/30 correct. Since it comes from the same test domain, it adds no evidence of performance on real road images.

3.2 Multi-Indicator and TOPSIS Results

The rankings broadly agree: ρ(AHP, Entropy) = 0.881, ρ(AHP, Combined) = 0.929, and ρ(Entropy, Combined) = 0.975. Perturbing each combined weight by ±20% keeps the Spearman correlation with the baseline at or above 0.997. For α = 0, 0.25, 0.5, 0.75, and 1.0, the correlations are 0.975, 0.991, 1.000, 0.996, and 0.929. Thus, the tested weight changes have relatively little effect on the ordering.

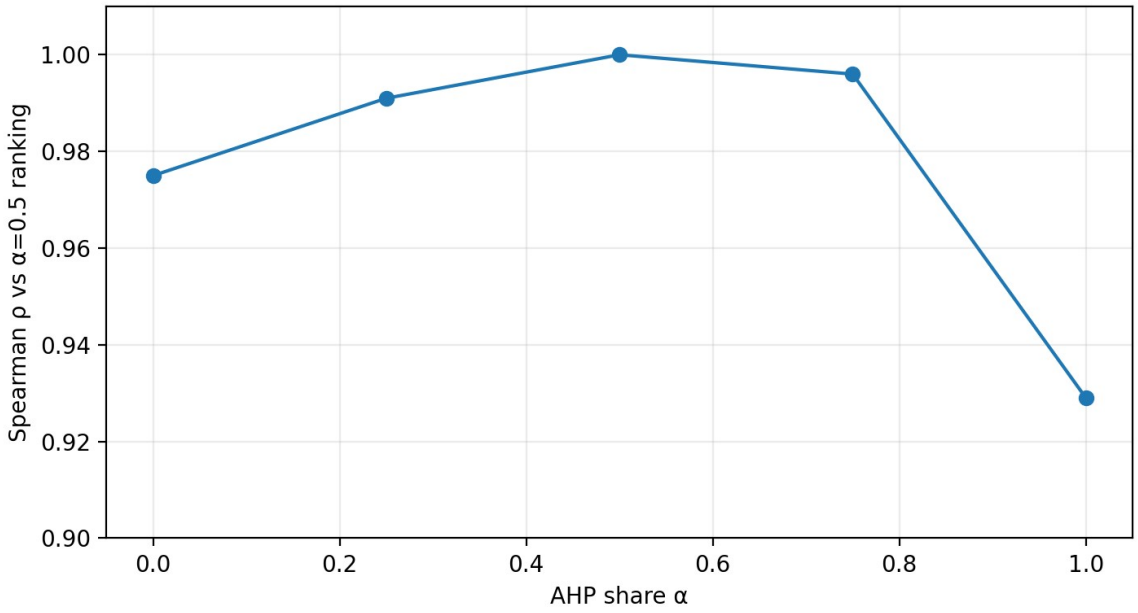


Figure 5. Sensitivity of the TOPSIS ranking to the AHP/entropy mixing ratio.

Edge Density receives an entropy weight of 0.8956, much larger than the other indicators. That weight comes from its normalized distribution in these images. It should not be read as saying that edges account for 89.56% of the physical fog effect. Stable rankings are useful, but neither ranking stability nor a large weight establishes accuracy against measured visibility.

3.3 ARIMA Proxy Forecast

AIC = 722.484 selects ARIMA(0,1,1). Its 20-point hold-out MAE is 16.376 km⁻¹ and RMSE is 19.536 km⁻¹. The illustrative 30-step forecast is almost flat at $\beta_{proxy} \approx 48.055$ km⁻¹, or about 81.41 m on the legacy scale. It stays below the legacy 150 m visibility threshold without a crossing. This describes the model output, not a verified future road condition: physical timing and continuity are still missing from the proxy sequence.

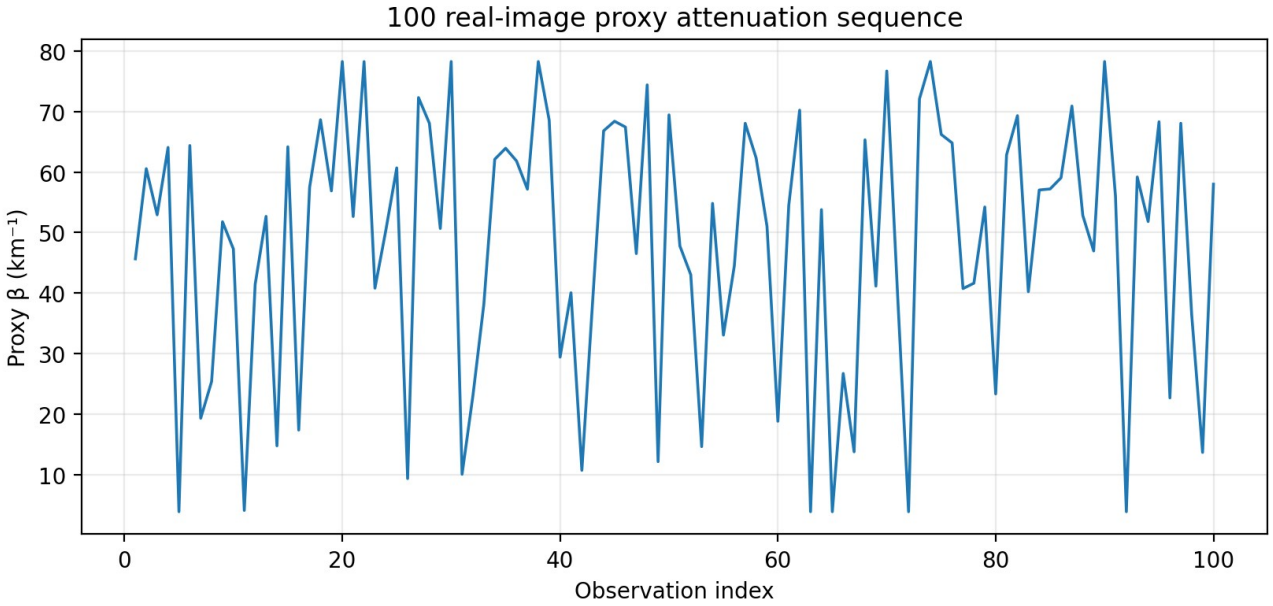


Figure 6. Existing 100-image proxy attenuation sequence. Observation index is not treated as physical time unless continuity is verified.

3.4 Forecasting Experimental Setup

Item	Specification
Training / test split	Observations 1–80 / 81–100
ARIMA candidates	p=0–3, d=0–2, q=0–3; selected by AIC
RF / XGBoost predictors	Previous three proxy β values
Shuffling	None
ML random seed	42
Metrics	MAE, RMSE, R^2, directional accuracy

Table 6. Common chronological model-comparison protocol.

3.5 ARIMA vs Random Forest vs XGBoost

Model	MAE	RMSE	R2	Directional Accuracy
ARIMA(0,1,1)	16.3759	19.5360	-0.0554	70.0%
Random Forest	17.3133	21.9277	-0.3296	70.0%
XGBoost	17.5265	22.5801	-0.4099	70.0%

Table 7. Formal hold-out comparison on the same proxy sequence.

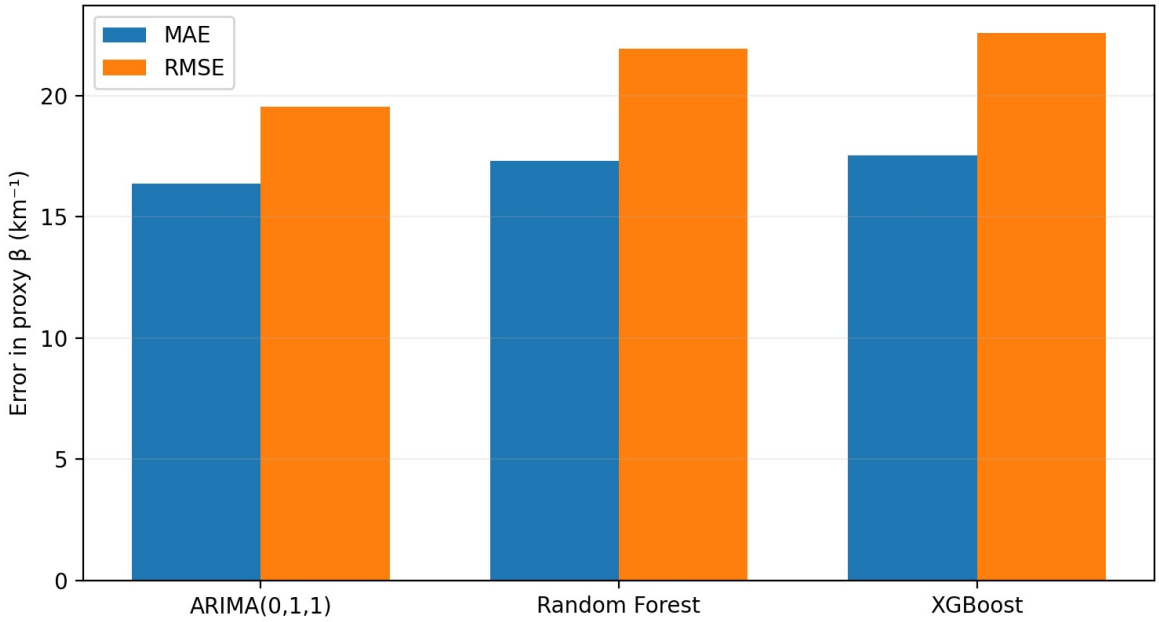


Figure 7. Hold-out MAE and RMSE. ARIMA has the lowest error among the three tested forecasting models.

ARIMA has the lowest errors in this comparison. Its MAE is about 5.4% below Random Forest and 6.6% below XGBoost; RMSE is about 10.9% and 13.5% lower, respectively. All three reach about 70% directional accuracy. Yet all have negative R^2, meaning their squared errors exceed those of a constant prediction at the hold-out target mean. That mean is a retrospective reference, not an available operational forecast. The result limits how much predictive success we can claim.

3.6 Integrated Result Interpretation

The model results answer different questions. CNN accuracy concerns synthetic density classification. Random Forest and XGBoost errors concern prediction of an image-derived proxy. The AMOS experiment asks whether a meteorological relationship transfers across events. Combining those numbers into one verdict on machine learning would hide the data and target differences that explain them.

AMOS makes the transfer problem particularly clear. Meteorological Random Forest reaches training $R^2 \approx 0.993$, but its independent-event R^2 falls below -1. A close fit to one event is therefore a poor guide to its next-event error. At the 5- and 15-minute horizons, temporal Ridge also has high R^2 but larger MAE than persistence. These comparisons set a concrete requirement for further development:

a future XGBoost, LSTM, or hybrid model should improve on a simple usable baseline when tested on unseen fog events.

Model / split	N	MAE (m)	RMSE (m)	R^2
Ridge - training day	1440	1681.23	2057.17	0.3876
Ridge - independent test day	1438	3125.61	4260.71	-0.4736
Random Forest - training day	1440	134.11	222.12	0.9929
Random Forest - independent test day	1438	4045.58	5119.92	-1.1278

Table 8. Independent-event AMOS meteorological validation demonstrates severe generalization loss despite excellent in-event Random Forest fit.

Horizon	Method	N	MAE (m)	RMSE (m)	R^2
5 min	Temporal Ridge	1410	515.95	784.34	0.9507
5 min	Persistence baseline	1410	154.01	430.42	0.9852
15 min	Temporal Ridge	1400	918.39	1374.65	0.8497
15 min	Persistence baseline	1400	299.43	720.31	0.9587

Table 9. AMOS short-term forecast comparison. Persistence outperforms the tested temporal Ridge model at both horizons.

In the highway sequence, smoothed contrast changes from about 0.09497 to 0.09605, an increase of only +1.13%. Raw contrast has a Spearman correlation of about 0.055 with frame order. These numbers provide little support for a strong, steadily dissipating fog trend. They describe relative image contrast; without geometric calibration, converting that small change into meters would add precision the recording cannot justify.

4. Discussion

4.1 How the Components Form One System

The components connect through the estimated state: images supply density and severity information, calibration would put them on a common scale, and a temporal model would forecast that state. The connection is useful to specify, even though it is not yet validated from end to end. Synthetic CNN samples and unlabeled road images cannot simply be pooled as if they shared measured β targets.

4.2 Role of Machine Learning in the Overall Framework

Each model has a limited job. The CNN learns to separate synthetic density classes. Random Forest and XGBoost test next-step prediction from three proxy lags. Meteorological Random Forest tests transfer between AMOS events. TOPSIS adds a way to inspect how image characteristics affect the severity ranking. This is especially helpful when an image score seems unexpected: its feature contributions can be examined. Such inspection helps explain a prediction, but does not replace comparison with independent measurements.

4.3 Why ARIMA Is Currently Best

We retain ARIMA as the leading proxy benchmark because it has the lowest measured hold-out errors here. One possible explanation is the small sample: there are only 100 observations, and the tree models receive just three lags. Flexible partitions may be difficult to estimate reliably under those conditions, while a low-order integrated moving-average model needs less structure. This explanation is consistent with the result, but the comparison does not isolate the cause. With one split, it also cannot establish that the difference will recur on another sequence.

Longer recordings and aligned meteorological variables could change this ordering. Random Forest, XGBoost, or sequence networks may then capture information absent from the present lag vector. That possibility needs rolling-origin or event-level tests, with attention to errors near low-visibility thresholds, rather than a claim of superiority based on model type.

4.4 Physical Meaning of β and Visibility

Extinction coefficient β links attenuation to visibility, but the conversion constant depends on the visibility definition. A 150 m threshold corresponds to $\beta \approx 19.97\ km^{-1}$ under the 5% convention and $\beta \approx 26.08\ km^{-1}$ under the 2% convention. The difference is substantial enough to affect threshold interpretation. We keep the existing proxy results on their original scale for reproducibility. Before using fused outputs for road decisions, one operational definition must be selected and every branch calibrated against it.

4.5 Model Selection and Future Improvement

The next dataset needs measurements that can connect the current experiments: consecutive road frames, precise timestamps, camera configuration, known or calibratable distances, visibility-meter readings, and weather variables. With those aligned, we could first test the image estimate, then calibrate CNN and TOPSIS to a shared β, and finally compare ARIMA/ARIMAX, tree models, and sequence networks using rolling-origin validation. Threshold-crossing error and low-visibility false negatives would then be meaningful targets. At present, adding model complexity alone would leave the main measurement gap unresolved.

- Split by event or camera so that neighboring frames do not make the test artificially easy.
- Report errors separately below 500 m, below 300 m, and near the chosen operational threshold.
- Check Softmax calibration before treating class probabilities as confidence.
- Align weather timestamps before adding exogenous variables to ARIMAX or tree models.
- Compare with persistence and other simple baselines on the same held-out targets.

4.6 Limitations

Limitation	Consequence
Synthetic-to-real domain gap	The 92.46% CNN accuracy is a synthetic-domain result, not real-road MOR accuracy.
No real-image ground truth	The 100-image feature/TOPSIS set lacks synchronized field visibility labels.
Temporal ambiguity	The proxy observation order is not independently verified as a continuous physical time series.
Convention mismatch	CNN and legacy proxy use different visibility constants; final fusion is not physically calibrated.
Small forecasting sample	Only 80 training and 20 test observations are available for ARIMA/RF/XGBoost comparison.
Highway scale missing	The independent highway sequence supports relative contrast trends but not absolute meter-scale MOR.
Limited AMOS events	Two event periods are insufficient to characterize seasonal and mechanism-level generalization.

Table 10. Main limitations that bound the interpretation of the reported results.

5. Conclusion

The experiments show progress in estimating image fog state, with more limited success in forecasting. The CNN reaches 92.46% synthetic test accuracy and Macro F1 = 0.9248. The five visual indicators produce TOPSIS rankings that are stable under the tested weight perturbations. On the 100-observation proxy sequence, ARIMA(0,1,1) has lower hold-out error than Random Forest and XGBoost, but all three have negative R^2. Thus, the evidence supports a synthetic classifier and a relative image ranking more strongly than a reliable physical forecast.

Two other findings shape the next step. The AMOS model that closely fits one event performs poorly on the second, and persistence beats temporal Ridge at both tested forecast horizons. The highway recording, meanwhile, shows only a small relative contrast change. Taken together, these results give us a modeling structure and specific problems to resolve. A road-visibility forecast still needs synchronized real video and independent measurements, verified temporal continuity, and a common β-to-visibility convention. Those are the conditions under which the proposed connections can be tested.

6. References

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[13] Supplied project materials (2026): CNN training/inference scripts, model weights, metrics JSON, training history, random-test records, AMOS data analysis, highway screenshot analysis, and the unified CNN–Feature–TOPSIS–ARIMA report.

Appendix A. Unified Equations

1. Image formation: $I_t(x) = J(x)\,\tau_t(x) + A_t[1 - \tau_t(x)]$.
2. Transmission: $\tau_t(x) = \exp[-\beta_t\, d(x)]$.
3. 5% visibility convention: $V_{5\%} = 2.995732/\beta$.
4. CNN continuous density: $\beta_{CNN,t} = 0.005P_{1,t} + 0.010P_{2,t} + 0.020P_{3,t}$ (m^{-1}).
5. Unit conversion for provisional fusion: $\beta_{CNN,t}\,(km^{-1}) = 1000\beta_{CNN,t}$.
6. Standardization: $Z(X_i) = (X_i - \mu_X)/\sigma_X$.
7. Legacy feature score: $S_{raw} = Z(Dark) + Z(Brightness) - Z(Contrast) - Z(Saturation) - Z(Edge)$.
8. Robust normalization: $S = \mathrm{clip}[(S_{raw} - P5)/(P95 - P5), 0, 1]$.
9. TOPSIS closeness: $C_i = D_i^- / (D_i^+ + D_i^-)$.
10. Legacy TOPSIS proxy: $\beta_{TOPSIS} = 3.912 + 74.328C_i$ km^{-1}.
11. Provisional common-state fusion: $\beta_{state,t} = \lambda\beta_{CNN,t}\,(km^{-1}) + (1-\lambda)\beta_{TOPSIS,t}$.
12. Legacy proxy visibility: $V_{proxy} = 3912/\beta_{proxy}$ (m).
13. RF/XGBoost lag vector: $x_t = [\beta_{t-1}, \beta_{t-2}, \beta_{t-3}]$.
14. Local threshold time in log-visibility form: $\Delta t_H = [\ln(H) - z_0]/g$, when direction and g permit a future crossing.

Appendix B. 100 Real-Image Proxy Observations

The values below are model-derived image features and proxy quantities, not measured atmospheric observations. The ordering is retained exactly for reproducibility, but it should not be interpreted as a verified physical time axis unless continuity is independently established.

Obs.	Image	Bright.	Contrast	Sat.	Dark	Edge	S	βproxy (km⁻¹)	Vproxy (m)
1	page01_001.png	0.696	45.71	0.032	0.685	0.200	0.562	45.66	85.7
2	page01_002.png	0.860	53.25	0.020	0.853	0.200	0.762	60.55	64.6
3	page01_003.png	0.798	54.51	0.028	0.788	0.200	0.659	52.93	73.9
4	page01_004.png	0.866	45.31	0.024	0.857	0.200	0.809	64.05	61.1
5	page01_005.png	0.365	74.15	0.419	0.301	0.200	0.000	3.91	1000.0
6	page01_006.png	0.867	46.21	0.018	0.859	0.200	0.813	64.37	60.8
7	page01_007.png	0.658	83.40	0.105	0.635	0.200	0.208	19.34	202.3
8	page01_008.png	0.701	74.78	0.125	0.671	0.200	0.289	25.41	153.9
9	page01_009.png	0.805	61.30	0.029	0.795	0.200	0.644	51.79	75.5
10	page01_010.png	0.772	61.50	0.029	0.762	0.200	0.584	47.29	82.7
11	page02_011.png	0.543	70.18	0.195	0.488	0.200	0.003	4.12	950.5
12	page02_012.png	0.704	57.05	0.036	0.691	0.200	0.505	41.46	94.4
13	page02_013.png	0.765	46.91	0.027	0.754	0.200	0.656	52.67	74.3
14	page02_014.png	0.639	79.73	0.145	0.598	0.200	0.146	14.79	264.6
15	page02_015.png	0.833	38.86	0.023	0.823	0.200	0.811	64.16	61.0
16	page02_016.png	0.666	79.21	0.149	0.623	0.200	0.181	17.39	225.0
17	page02_017.png	0.771	38.55	0.022	0.762	0.200	0.721	57.48	68.1
18	page02_018.png	0.844	38.15	0.014	0.838	0.200	0.871	68.63	57.0
19	page02_019.png	0.813	45.29	0.043	0.795	0.200	0.713	56.88	68.8
20	page02_020.png	0.888	32.67	0.008	0.885	0.199	1.000	78.24	50.0
21	page03_021.png	0.729	39.02	0.024	0.720	0.200	0.656	52.65	74.3
22	page03_022.png	0.850	42.92	0.013	0.845	0.198	1.000	78.23	50.0
23	page03_023.png	0.697	47.80	0.073	0.672	0.200	0.497	40.82	95.8
24	page03_024.png	0.757	46.96	0.042	0.742	0.200	0.625	50.39	77.6
25	page03_025.png	0.783	35.33	0.015	0.777	0.200	0.764	60.68	64.5
26	page03_026.png	0.534	78.78	0.095	0.511	0.200	0.074	9.38	417.0
27	page03_027.png	0.866	44.82	0.015	0.860	0.199	0.920	72.29	54.1
28	page03_028.png	0.895	45.36	0.016	0.889	0.200	0.863	68.03	57.5
29	page03_029.png	0.773	71.28	0.019	0.769	0.199	0.629	50.68	77.2
30	page03_030.png	0.907	38.73	0.008	0.904	0.195	1.000	78.24	50.0
31	page04_031.png	0.612	72.44	0.194	0.552	0.200	0.083	10.09	387.8
32	page04_032.png	0.683	75.64	0.120	0.643	0.200	0.259	23.13	169.1
33	page04_033.png	0.643	49.15	0.041	0.629	0.200	0.461	38.18	102.5
34	page04_034.png	0.817	40.71	0.017	0.810	0.200	0.783	62.10	63.0
35	page04_035.png	0.839	40.58	0.017	0.832	0.200	0.807	63.92	61.2
36	page04_036.png	0.835	47.20	0.019	0.828	0.200	0.779	61.82	63.3
37	page04_037.png	0.842	53.36	0.035	0.830	0.200	0.716	57.16	68.4
38	page04_038.png	0.849	61.96	0.000	0.849	0.198	1.000	78.24	50.0
39	page04_039.png	0.841	30.58	0.016	0.835	0.200	0.870	68.55	57.1
40	page04_040.png	0.736	73.31	0.121	0.695	0.200	0.343	29.43	132.9
41	page05_041.png	0.737	63.97	0.054	0.719	0.200	0.486	40.06	97.7
42	page05_042.png	0.631	81.86	0.172	0.588	0.200	0.092	10.74	364.2
43	page05_043.png	0.754	69.25	0.062	0.737	0.200	0.471	38.94	100.5
44	page05_044.png	0.874	50.48	0.021	0.866	0.200	0.846	66.80	58.6
45	page05_045.png	0.883	42.22	0.013	0.877	0.200	0.867	68.38	57.2
46	page05_046.png	0.865	40.12	0.012	0.861	0.200	0.854	67.42	58.0
47	page05_047.png	0.727	48.27	0.050	0.712	0.200	0.573	46.53	84.1
48	page05_048.png	0.922	36.78	0.016	0.916	0.200	0.948	74.38	52.6
49	page05_049.png	0.611	75.38	0.159	0.563	0.200	0.111	12.18	321.1
50	page05_050.png	0.860	34.28	0.011	0.855	0.200	0.882	69.43	56.3
51	page06_051.png	0.764	56.00	0.038	0.751	0.200	0.590	47.79	81.9
52	page06_052.png	0.741	59.05	0.050	0.726	0.200	0.526	43.04	90.9
53	page06_053.png	0.677	78.35	0.212	0.628	0.200	0.145	14.66	266.9
54	page06_054.png	0.830	55.92	0.035	0.818	0.200	0.685	54.81	71.4

Obs.	Image	Bright.	Contrast	Sat.	Dark	Edge	S	βproxy (km⁻¹)	Vproxy (m)
55	page06_055.png	0.699	65.26	0.082	0.676	0.200	0.392	33.07	118.3
56	page06_056.png	0.766	59.23	0.060	0.746	0.200	0.546	44.50	87.9
57	page06_057.png	0.876	38.92	0.023	0.868	0.200	0.863	68.04	57.5
58	page06_058.png	0.873	48.97	0.035	0.862	0.200	0.786	62.33	62.8
59	page06_059.png	0.794	55.55	0.037	0.781	0.200	0.634	51.04	76.6
60	page06_060.png	0.669	79.55	0.140	0.639	0.200	0.201	18.84	207.6
61	page07_061.png	0.845	55.73	0.056	0.826	0.200	0.681	54.49	71.8
62	page07_062.png	0.894	44.56	0.013	0.888	0.200	0.892	70.21	55.7
63	page07_063.png	0.505	69.53	0.202	0.452	0.200	0.000	3.91	1000.0
64	page07_064.png	0.829	64.52	0.018	0.826	0.200	0.671	53.78	72.7
65	page07_065.png	0.443	74.62	0.311	0.382	0.200	0.000	3.91	1000.0
66	page07_066.png	0.710	74.12	0.117	0.669	0.200	0.307	26.73	146.3
67	page07_067.png	0.581	66.78	0.156	0.548	0.200	0.133	13.80	283.4
68	page07_068.png	0.856	46.05	0.018	0.849	0.200	0.826	65.31	59.9
69	page07_069.png	0.693	54.82	0.036	0.680	0.200	0.501	41.16	95.1
70	page07_070.png	0.925	32.64	0.012	0.920	0.200	0.979	76.67	51.0
71	page08_071.png	0.663	48.78	0.048	0.647	0.200	0.479	39.48	99.1
72	page08_072.png	0.520	86.62	0.264	0.471	0.200	0.000	3.91	1000.0
73	page08_073.png	0.881	43.25	0.015	0.874	0.199	0.917	72.07	54.3
74	page08_074.png	0.782	68.93	0.027	0.775	0.196	1.000	78.24	50.0
75	page08_075.png	0.817	43.02	0.022	0.808	0.199	0.838	66.20	59.1
76	page08_076.png	0.820	56.16	0.032	0.810	0.199	0.819	64.80	60.4
77	page08_077.png	0.735	57.23	0.074	0.711	0.200	0.496	40.75	96.0
78	page08_078.png	0.731	57.18	0.063	0.712	0.200	0.508	41.64	94.0
79	page08_079.png	0.794	48.53	0.036	0.782	0.200	0.677	54.22	72.2
80	page08_080.png	0.679	73.95	0.128	0.651	0.200	0.262	23.35	167.5
81	page09_081.png	0.853	45.74	0.021	0.844	0.200	0.793	62.83	62.3
82	page09_082.png	0.901	44.19	0.016	0.894	0.200	0.880	69.30	56.5
83	page09_083.png	0.696	54.60	0.054	0.679	0.200	0.489	40.24	97.2
84	page09_084.png	0.799	49.64	0.017	0.794	0.200	0.715	57.03	68.6
85	page09_085.png	0.768	39.47	0.018	0.761	0.200	0.717	57.20	68.4
86	page09_086.png	0.844	50.40	0.032	0.833	0.200	0.742	59.04	66.3
87	page09_087.png	0.883	38.51	0.017	0.875	0.200	0.901	70.89	55.2
88	page09_088.png	0.795	52.80	0.036	0.784	0.200	0.658	52.83	74.1
89	page09_089.png	0.747	53.14	0.041	0.734	0.200	0.579	46.96	83.3
90	page09_090.png	0.937	31.76	0.007	0.934	0.200	1.000	78.24	50.0
91	page10_091.png	0.787	46.31	0.019	0.780	0.200	0.701	56.03	69.8
92	page10_092.png	0.528	88.65	0.290	0.475	0.200	0.000	3.91	1000.0
93	page10_093.png	0.791	41.42	0.022	0.782	0.200	0.744	59.18	66.1
94	page10_094.png	0.773	50.71	0.029	0.763	0.200	0.645	51.82	75.5
95	page10_095.png	0.877	46.13	0.015	0.871	0.200	0.866	68.29	57.3
96	page10_096.png	0.657	73.51	0.107	0.623	0.200	0.253	22.70	172.4
97	page10_097.png	0.875	39.46	0.018	0.867	0.200	0.863	68.04	57.5
98	page10_098.png	0.708	60.61	0.077	0.686	0.200	0.437	36.42	107.4
99	page10_099.png	0.611	72.59	0.155	0.563	0.200	0.132	13.70	285.6
100	page10_100.png	0.843	53.34	0.028	0.833	0.200	0.728	57.99	67.5

Appendix C. Reproducible Model-Comparison Protocol

1. Preserve chronological order; do not randomly shuffle the proxy sequence.

2. Use observations 1–80 for training and 81–100 for testing.

3. Select ARIMA order only from the training portion using AIC over p=0–3, d=0–2, q=0–3.

4. Build Random Forest and XGBoost predictors only from the previous three β values available at prediction time.

5. Use the same 20 test targets for every model and random seed 42 for machine-learning reproducibility.

6. Report MAE, RMSE, R^2, and directional accuracy. Do not hide negative R^2.

7. Do not interpret the proxy benchmark as a validated physical road forecast unless sequence continuity and field labels are independently verified.

8. For future continuous field data, replace one fixed hold-out split with rolling-origin or event-level validation and add meteorological exogenous variables only after timestamp alignment is verified.